Retail Intelligence Lab — Consolidated Data Warehouse & Mining Report

Course: DTU SE425 — Data Warehouse and Data Mining | Generated: 2026-10-26 18:19:54 UTC

Metric	Value
Pipeline Status	COMPLETED
Completed Stages	18 / 18
Data Warehouse	SQLite 3 (Star & Snowflake Schema)
Primary Dataset	Online Retail.csv (541k records)

Unit 6 — Complex Data Mining Implementation Matrix

Unit	Topic	Implementation Method	Verification Status
6A	Multidimensional Analysis	Relational Star/Snowflake OLAP	Implemented & Verified
6B	Spatial Database Mining	Haversine Distance + Spatial DBSCAN	Implemented & Verified
6C	Multimedia Database Mining	MD5 Exact + pHash/dHash Perceptual Duplicates	Implemented & Verified
6D	Time-Series & Sequence Mining	Weekly Resampling, Naive/MA/Trend Forecast + Sequential Itemset Analysis	Implemented & Verified
6E	Text Database Mining	TF-IDF + Cosine Search + Description Clustering	Implemented & Verified
6F	Applications and Trends	Retail Domain Synthesis & Educational Trends Analysis	Implemented & Verified

Conclusions & Real Findings

Real transactional retail data demonstrates strong customer clustering by monetary frequency, distinct product associations among homeware gift items, and clear seasonal trends. Pruned decision trees effectively predict repeat purchase while preventing target leakage.

Academic Methodology & Limitations

The dataset lacks natural GPS coordinates (handled safely via explicit spatial coordinate upload), has unrecorded guest checkout demographics, and exhibits significant positive skew in transaction quantities.